

Jacobian Diagnostics for ABBA4 Implicit

Three ideal-root tangents along one signed triple jump

Physical GC2D trajectory, local symplecticity, and accumulated sensitivity

1 Purpose of the diagnostics

An `ABBA4Implicit` integration can be observed through the physical state and an infinitesimal perturbation,

$$z_n \in \mathbb{R}^2, \quad \xi_n \in \mathbb{R}^2.$$

The trajectory advances with the complete three-solve method,

$$z_{n+1} = \Psi_{h,t_n}^{[4]}(z_n),$$

while the tangent follows the derivative of that same nonlinear map,

$$\xi_{n+1} = D\Psi_{h,t_n}^{[4]}(z_n) \xi_n.$$

The Jacobian is not a replacement for the state update. In general,

$$z_{n+1} \neq D\Psi_{h,t_n}^{[4]}(z_n) z_n.$$

It acts on a perturbation or on an accumulated derivative with respect to the initial condition.

The production observer exposes the complete accepted outer step and an ordered tuple of three signed substep snapshots. The diagnostic first constructs one ideal-root physical Jacobian from each snapshot and then applies the chain rule.

2 Signed substep geometry

Let

$$(c_1, c_2, c_3) = (\gamma, \delta, \gamma),$$

where

$$\gamma = \frac{1}{2 - 2^{1/3}}, \quad \delta = -\frac{2^{1/3}}{2 - 2^{1/3}}.$$

Define

$$\ell_j = c_j h, \quad \tau_j = t_n + h \sum_{r < j} c_r, \quad s_j = \frac{\ell_j}{2}.$$

Then

$$z^{[0]} = z_n, \quad z^{[j]} = \Psi_{\ell_j, \tau_j}^{[2]}(z^{[j-1]}), \quad z_{n+1} = z^{[3]}.$$

The signed clock is

Substep	Start	Duration	End
1	t_n	γh	$t_n + \gamma h$
2	$t_n + \gamma h$	δh	$t_n + (\gamma + \delta)h$
3	$t_n + (\gamma + \delta)h$	γh	$t_n + h$

The second row runs backward in time. A diagnostic must use the stored signed duration and start time rather than infer monotone stage times from the outer interval.

3 One projected ABBA substep

For a provisional multiplier $\mu_j \in \mathbb{R}^2$, initialize

$$u_j^{(0)} = z^{[j-1]} + \mu_j, \quad v_j^{(0)} = z^{[j-1]} - \mu_j.$$

The endpoint-time stages are

$$\begin{aligned} u_j^{(1)} &= u_j^{(0)} + s_j f(v_j^{(0)}, \tau_j), \\ v_j^{(1)} &= v_j^{(0)} + s_j f(u_j^{(1)}, \tau_j), \\ v_j^{(2)} &= v_j^{(1)} + s_j f(u_j^{(1)}, \tau_j + \ell_j), \\ u_j^{(2)} &= u_j^{(1)} + s_j f(v_j^{(2)}, \tau_j + \ell_j). \end{aligned}$$

The independent root μ_j^* satisfies

$$R_j(z^{[j-1]}, \mu_j) = u_j^{(2)} - v_j^{(2)} + 2\mu_j = 0.$$

At the exact root,

$$z^{[j]} = u_j^{(2)} + \mu_j^* = v_j^{(2)} - \mu_j^*.$$

The finite-tolerance implementation returns the neutral mean of these corrected copies. The analytic formulas below evaluate the ideal-root tangent at the converged stages; a finite-difference derivative of the emitted map also contains the stopping decision and the residual-level copy mismatch.

4 Small GC2D matrices at the traversed stages

For

$$f(z, t) = J_0 \nabla_z \phi(z, t), \quad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

introduce the signed small matrix

$$w_j(z, t) := s_j D_z f(z, t) = s_j \begin{pmatrix} -\phi_{XY} & -\phi_{YY} \\ \phi_{XX} & \phi_{XY} \end{pmatrix}_{(z,t)}.$$

The four matrices for substep j are

$$\begin{aligned} w_{j,1} &= w_j(v_j^{(0)}, \tau_j), & w_{j,2} &= w_j(u_j^{(1)}, \tau_j), \\ w_{j,3} &= w_j(u_j^{(1)}, \tau_j + \ell_j), & w_{j,4} &= w_j(v_j^{(2)}, \tau_j + \ell_j), \end{aligned}$$

with

$$\Sigma_j = w_{j,2} + w_{j,3}.$$

Because $s_2 < 0$, all four middle-substep matrices include that sign. Replacing them by absolute-step matrices differentiates a different map.

5 Four duplicated-space stage Jacobians

Each elementary shear has a 4×4 Jacobian containing one small block:

$$\mathcal{W}_{j,1} = \begin{pmatrix} I_2 & w_{j,1} \\ 0 & I_2 \end{pmatrix}, \quad \mathcal{W}_{j,2} = \begin{pmatrix} I_2 & 0 \\ w_{j,2} & I_2 \end{pmatrix},$$

$$\mathcal{W}_{j,3} = \begin{pmatrix} I_2 & 0 \\ w_{j,3} & I_2 \end{pmatrix}, \quad \mathcal{W}_{j,4} = \begin{pmatrix} I_2 & w_{j,4} \\ 0 & I_2 \end{pmatrix}.$$

The exact Jacobian of the unprojected signed ABBA map is the ordered product

$$D\Phi_{\ell_j, \tau_j}^{\text{ABBA}} = \mathcal{W}_{j,4} \mathcal{W}_{j,3} \mathcal{W}_{j,2} \mathcal{W}_{j,1}.$$

Writing it as

$$D\Phi_{\ell_j, \tau_j}^{\text{ABBA}} = \begin{pmatrix} A_j & B_j \\ C_j & D_j \end{pmatrix},$$

the blocks are

$$\begin{aligned} A_j &= I_2 + w_{j,4} \Sigma_j, \\ B_j &= w_{j,1} + w_{j,4} + w_{j,4} \Sigma_j w_{j,1}, \\ C_j &= \Sigma_j, \\ D_j &= I_2 + \Sigma_j w_{j,1}. \end{aligned}$$

6 Local physical Jacobian

Form the four 2×2 matrices

$$\begin{aligned} K_j &= 4I_2 - (w_{j,1} + w_{j,2} + w_{j,3} + w_{j,4}) \\ &\quad + (w_{j,4} \Sigma_j + \Sigma_j w_{j,1}) - w_{j,4} \Sigma_j w_{j,1}, \\ L_j &= w_{j,1} - w_{j,2} - w_{j,3} + w_{j,4} \\ &\quad + (w_{j,4} \Sigma_j - \Sigma_j w_{j,1}) + w_{j,4} \Sigma_j w_{j,1}, \\ P_j &= I_2 + w_{j,1} + w_{j,4} + w_{j,4} \Sigma_j + w_{j,4} \Sigma_j w_{j,1}, \\ Q_j &= 2I_2 - w_{j,1} - w_{j,4} + w_{j,4} \Sigma_j - w_{j,4} \Sigma_j w_{j,1}. \end{aligned}$$

Here

$$K_j = D_{\mu_j} R_j, \quad L_j = D_{z^{[j-1]}} R_j.$$

The implicit-function theorem gives

$$D\mu_j^* = -K_j^{-1} L_j,$$

and the local physical Jacobian is

$$J_j := D\Psi_{\ell_j, \tau_j}^{[2]}(z^{[j-1]}) = P_j - Q_j K_j^{-1} L_j.$$

The matrix K_j is the reduced Newton matrix, not the physical Jacobian. To construct J_j without an inverse, solve

$$K_j T_j = L_j, \quad J_j = P_j - Q_j T_j.$$

7 Applying the triple-jump Jacobian

The complete outer Jacobian is

$$\mathcal{J}_n := D\Psi_{h, t_n}^{[4]}(z_n) = J_3 J_2 J_1.$$

If only one tangent vector is needed, no $\mathcal{J}_n$ need be formed. Start with $\eta_0 = \xi_n$ and repeat

$$b_j = L_j \eta_{j-1}, \quad K_j y_j = b_j, \quad \eta_j = P_j \eta_{j-1} - Q_j y_j, \quad j = 1, 2, 3.$$

Then

$$\xi_{n+1} = \eta_3.$$

This procedure uses three 2×2 linear solves per particle but introduces no additional nonlinear solve.

An equivalent local construction differentiates the four vector-field increments. With $M_j = D\mu_j^*$,

$$\begin{aligned} Dk_{j,1} &= \frac{w_{j,1}}{s_j}(I_2 - M_j), \\ Du_j^{(1)} &= I_2 + M_j + s_j Dk_{j,1}, \\ Dk_{j,2} &= \frac{w_{j,2}}{s_j} Du_j^{(1)}, \\ Dk_{j,3} &= \frac{w_{j,3}}{s_j} Du_j^{(1)}, \\ Dv_j^{(2)} &= I_2 - M_j + s_j(Dk_{j,2} + Dk_{j,3}), \\ Dk_{j,4} &= \frac{w_{j,4}}{s_j} Dv_j^{(2)}, \end{aligned}$$

and

$$J_j = I_2 + \frac{\ell_j}{4}(Dk_{j,1} + Dk_{j,2} + Dk_{j,3} + Dk_{j,4}).$$

This recurrence is useful as an independent analytic cross-check of the implicit-function factorization.

8 Observer-to-formula correspondence

One accepted `ABBA4ImplicitIntegrationStep` contains the outer state, time interval, coefficients, aggregate solver data, and an ordered `substeps` tuple. Each substep record supplies:

- signed `start_time` and `duration`;
- `state_before` and `state_after`;
- the converged multiplier;
- $u_j^{(0)}, v_j^{(0)}, u_j^{(1)}, v_j^{(2)}, u_j^{(2)}$;
- formulation, solver, iteration, residual, and tolerance metadata; and
- the dynamics object needed to evaluate the four spatial Jacobians.

The three snapshots are continuous:

$$z^{[0]} = z_n, \quad z_{\text{after}}^{[j]} = z_{\text{before}}^{[j+1]}, \quad z^{[3]} = z_{n+1}.$$

Shadow compositions used only to produce off-grid output samples do not emit observation records and must not enter an accumulated tangent diagnostic.

9 One-step diagnostic workflow

For each observed outer step:

1. Verify that exactly three substep records are present and that their coefficients, signed times, and boundary states are continuous.
2. For each j , evaluate $w_{j,1}, \dots, w_{j,4}$ at the stored stage points and signed endpoint times.

3. Form K_j, L_j, P_j, Q_j in the stated order and solve the required local 2×2 systems.
4. Propagate the tangent chronologically through J_1, J_2 , and J_3 , or construct $\mathcal{J}_n = J_3 J_2 J_1$.
5. Compare the implicit-function and stage-increment factorizations when diagnosing analytic derivatives.
6. Optionally compare with centered finite differences of the complete stored `map_state`, choosing a perturbation above the nonlinear residual floor.
7. Record local symplecticity and determinant defects without modifying the accepted physical trajectory.

10 Accumulated sensitivity

To differentiate the numerical solution with respect to its initial state, propagate $M_n \in \mathbb{R}^{2 \times 2}$:

$$M_0 = I_2, \quad \boxed{M_{n+1} = \mathcal{J}_n M_n.}$$

After N outer steps,

$$M_N = \mathcal{J}_{N-1} \cdots \mathcal{J}_1 \mathcal{J}_0 = \frac{\partial z_N}{\partial z_0}.$$

The newest Jacobian multiplies from the left. The update can be implemented by applying the three local factorizations successively to the two columns of M_n .

11 Symplecticity diagnostics

At exact projected roots, every local physical tangent is symplectic:

$$J_j^T J_0 J_j = J_0.$$

Therefore,

$$\mathcal{J}_n^T J_0 \mathcal{J}_n = J_0, \quad M_N^T J_0 M_N = J_0.$$

In two dimensions this is equivalent to unit determinant. Useful monitored defects are

$$\varepsilon_{\text{sp},n} = \left\| \mathcal{J}_n^T J_0 \mathcal{J}_n - J_0 \right\|, \quad \varepsilon_{\text{det},n} = |\det \mathcal{J}_n - 1|.$$

Their numerical size includes floating-point error, derivative accuracy, and the residuals of all three nonlinear solves. A small outer defect can mask opposing local defects, so diagnostics should retain both the three J_j values and their product when investigating a failure.

12 Nearby-trajectory interpretation

Let

$$z_n^\varepsilon = z_n + \varepsilon \xi_n.$$

Differentiability of the complete implicit map gives

$$\Psi_{h,t_n}^{[4]}(z_n^\varepsilon) = \Psi_{h,t_n}^{[4]}(z_n) + \varepsilon \mathcal{J}_n \xi_n + \mathcal{O}(\varepsilon^2).$$

Thus the integrator produces the reference trajectory while the optional diagnostic produces the linear evolution of an infinitesimal separation. The negative middle duration changes the intermediate tangent, but the three chronological applications still approximate the final nearby-trajectory difference.

13 Implementation-ready summary

For every outer step, perform

$z_{n+1} = \Psi_{h,t_n}^{[4]}(z_n) \quad \text{with three independent projected solves,}$ $\xi_{n+1} = J_3 J_2 J_1 \xi_n \quad \text{with three local ideal-root tangents.}$
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For each substep, compute

$$K_j y_j = L_j \eta_{j-1}, \quad \eta_j = P_j \eta_{j-1} - Q_j y_j.$$

The physical state is never obtained by multiplying its previous value by a Jacobian. The tangent is a read-only diagnostic derived from the already accepted multipliers and stage snapshots.

Minimum retained data

A reproducible outer-step record must preserve the ordered coefficient vector, the three signed start times and durations, the three accepted boundary states, and each substep's converged multiplier and stage snapshots. Keeping only the aggregate iteration count or the maximum multiplier norm is insufficient to reconstruct J_1 , J_2 , and J_3 .

Interpreting discrepancies

When two Jacobian diagnostics disagree, check the following causes in order:

- a lost negative sign in $s_2 = \delta h/2$ or a stage evaluated at the outer time instead of its signed local endpoint;
- reordered matrix products, especially $w_{j,4} \Sigma_j w_{j,1}$ or the outer product $J_3 J_2 J_1$;
- comparison of an ideal-root analytic tangent with a finite-difference perturbation below the nonlinear residual floor;
- an ill-conditioned K_j , which amplifies residual, derivative, and round-off errors; or
- accidental inclusion of an unobserved shadow composition in the accumulated tangent history.

These checks localize the defect to one signed solve before the complete outer composition is assessed.